



# Breach, Log, Renew

## Logging a Shared Robot Mower’s Virtual-Fence Breaches for Committee Lease Review

Lindiwe Achterberg · Solveig Adeyemi — School of Emergent Priorities

### The problem

Shared robot mowers increasingly serve community gardens, yet virtual-fence excursions into raised beds are common and rarely traced to a cause. Little is known about whether a committee-reviewed containment ledger changes lease-renewal decisions.

### Research questions

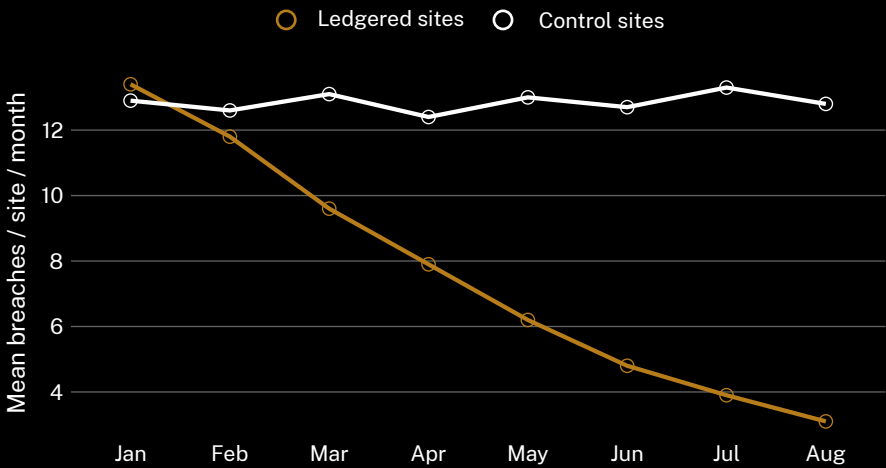
- Whether logged breach counts predict committee non-renewal at lease review
- Whether ledger visibility changes plot-holder trust in the mower
- Which virtual-fence geometries produce the most excursions per bed-hour

### Approach

- 22 mower-sites (14 ledgered, 8 control) · RTK-GPS boundary telemetry · 5-min polling · 8-month deployment
- Mixed-effects model of lease non-renewal on cumulative logged breaches
- Fence geometry and firmware version held constant across sites; thresholds pre-registered

### Findings

Mean monthly breaches at ledgered sites fall from double to low single figures across the deployment window. Non-renewal correlates with breach count only where the ledger is visible to the full committee ( $r = 0.52$ ). Firmware and fence-geometry updates over the same window produce no comparable shift at the unlogged control sites.



Mean monthly virtual-fence breaches at ledgered sites fell from 13.4 to 3.1 between January and August 2025 ( $n = 14$  mower-sites); 8 unlogged control sites showed no comparable decline.

### Implications

Visibility, not detection, appears to drive compliance: the mower does not breach less because it is logged, but because someone is now known to be watching. A ledgered breach becomes a lease-relevant fact; an unlogged one remains gossip, and the decline tracks the ledger’s introduction rather than any change to the mower itself. The effect resembles the broader observation-changes-behaviour literature, applied here to a lawnmower rather than a ward round. Next: pre-register a non-renewal threshold before a committee sees the ledger, not after.

### Selected references

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Boundary telemetry retained 18 months; no plot-holder identified; committee access to the ledger logged in turn.

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